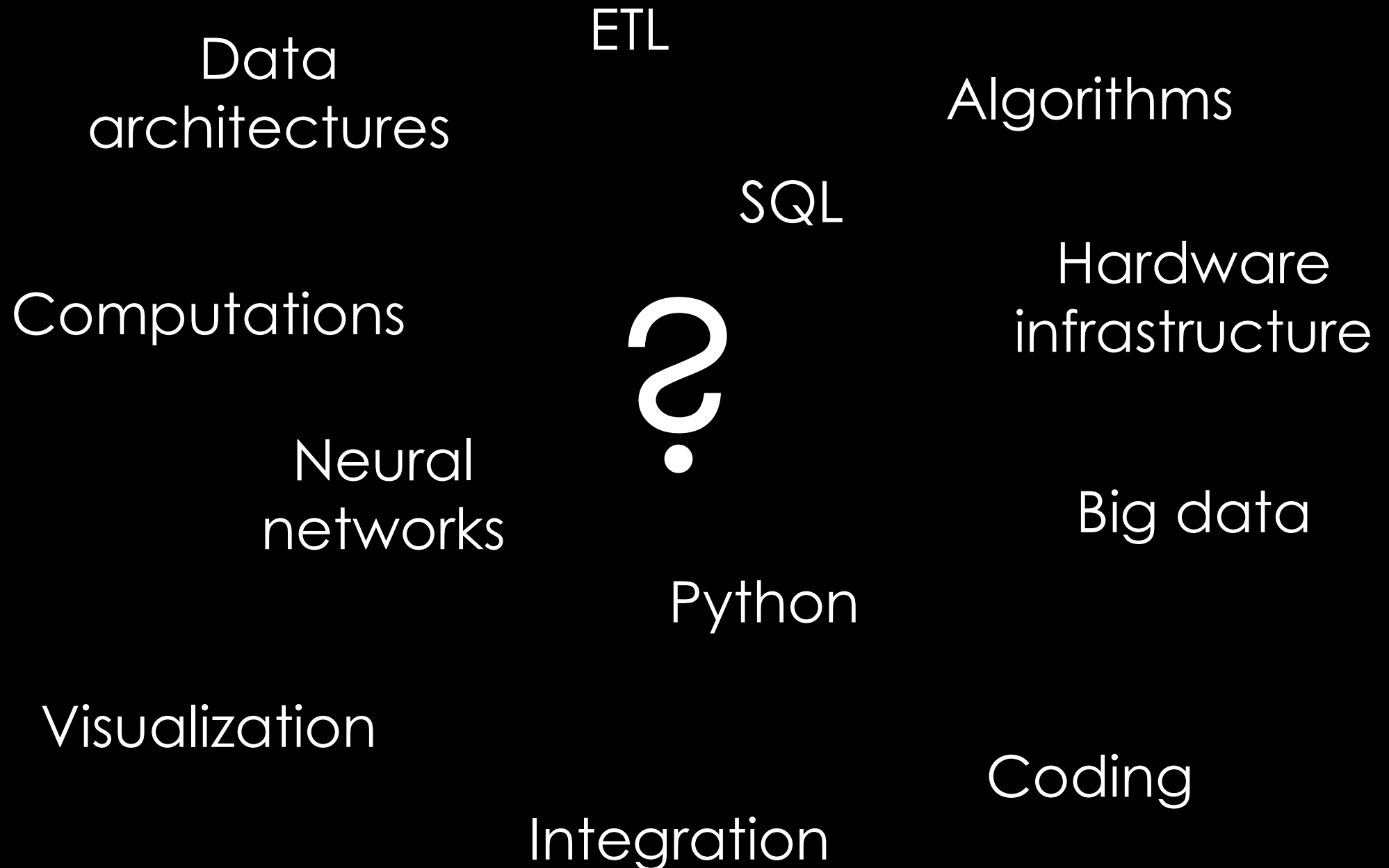


Introduction

Data Science Lab – 2025/26

What is «data science»?



What is «data science»

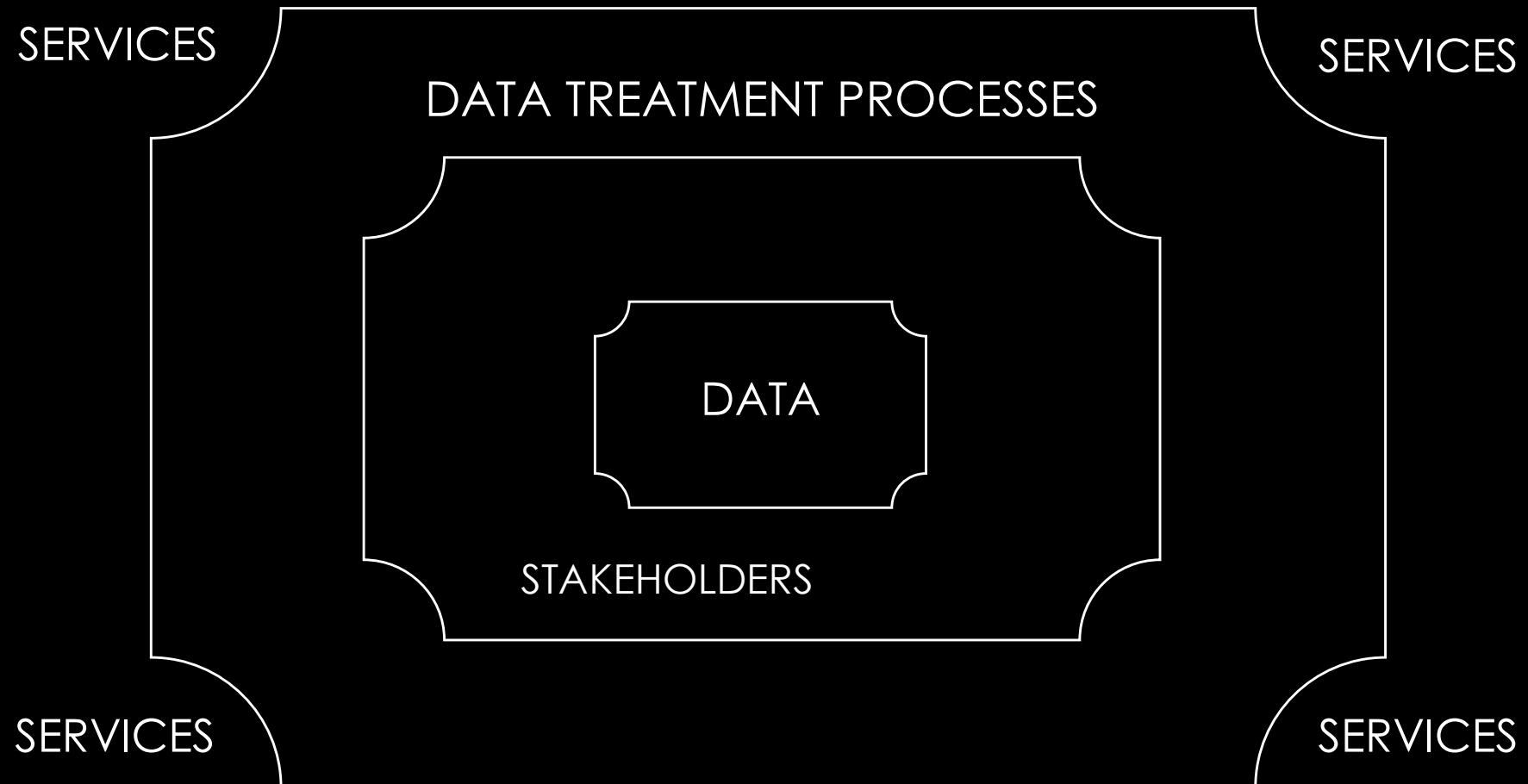
Reality → Data → Treatment → Stakeholder

A data science process is well-designed and effective when it **reveals something true** about the trait or fact of interest. Technically speaking, the key issue is its **validity**.

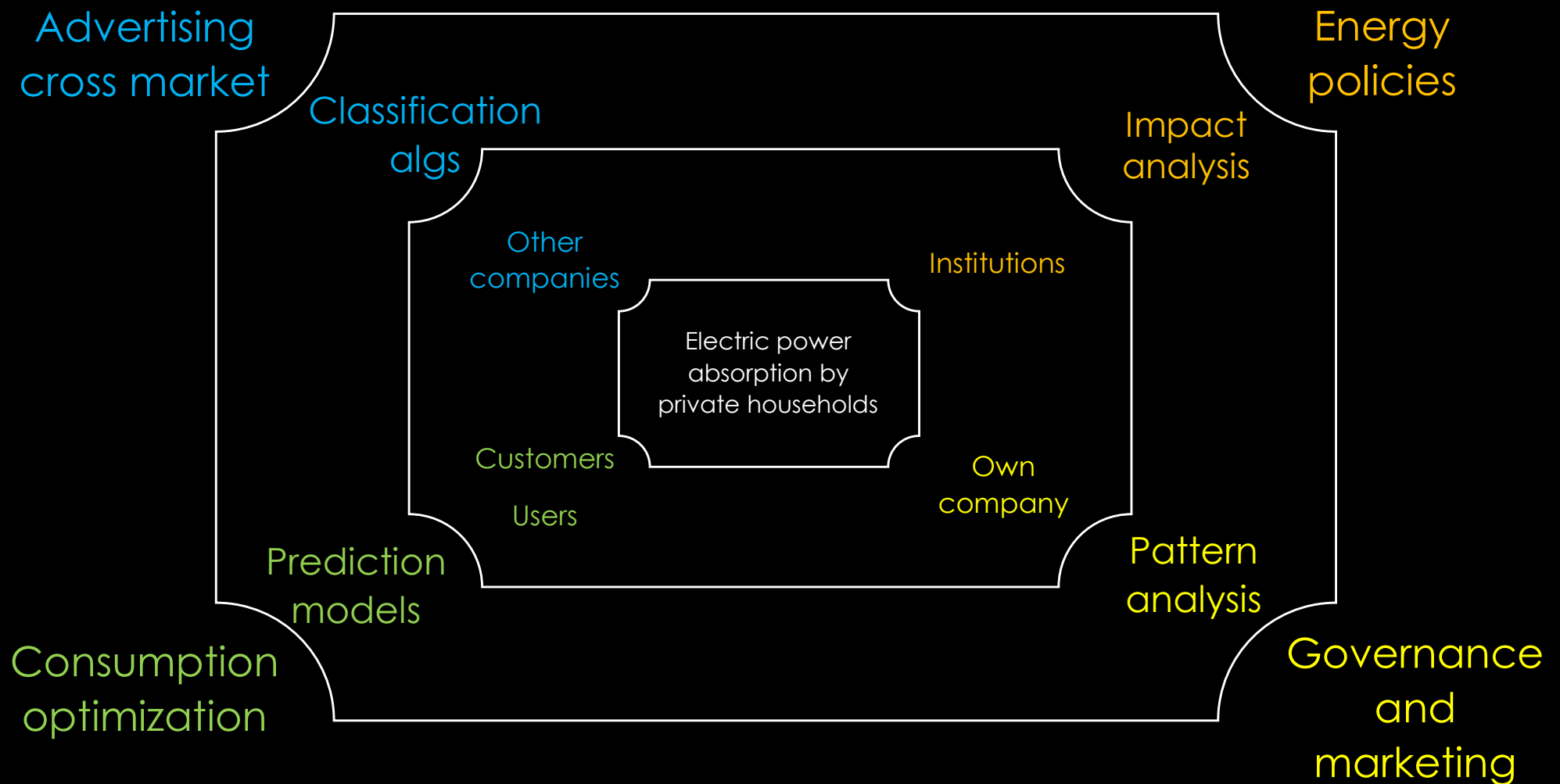
So the criterion you have to follow, when «doing» data science, is to take care that any component of the process is well-designed, dose not introduce biases or distortions and is properly integrated with the others.

And since some distortions are unavaoidable, you must be aware of them and to what extent the process is **valid**.

Part of a system



Example



Data science and statistics

The discipline whose aim is to work out the criteria and the methodologies assuring the validity of an «**informational**» process it **statistics**.

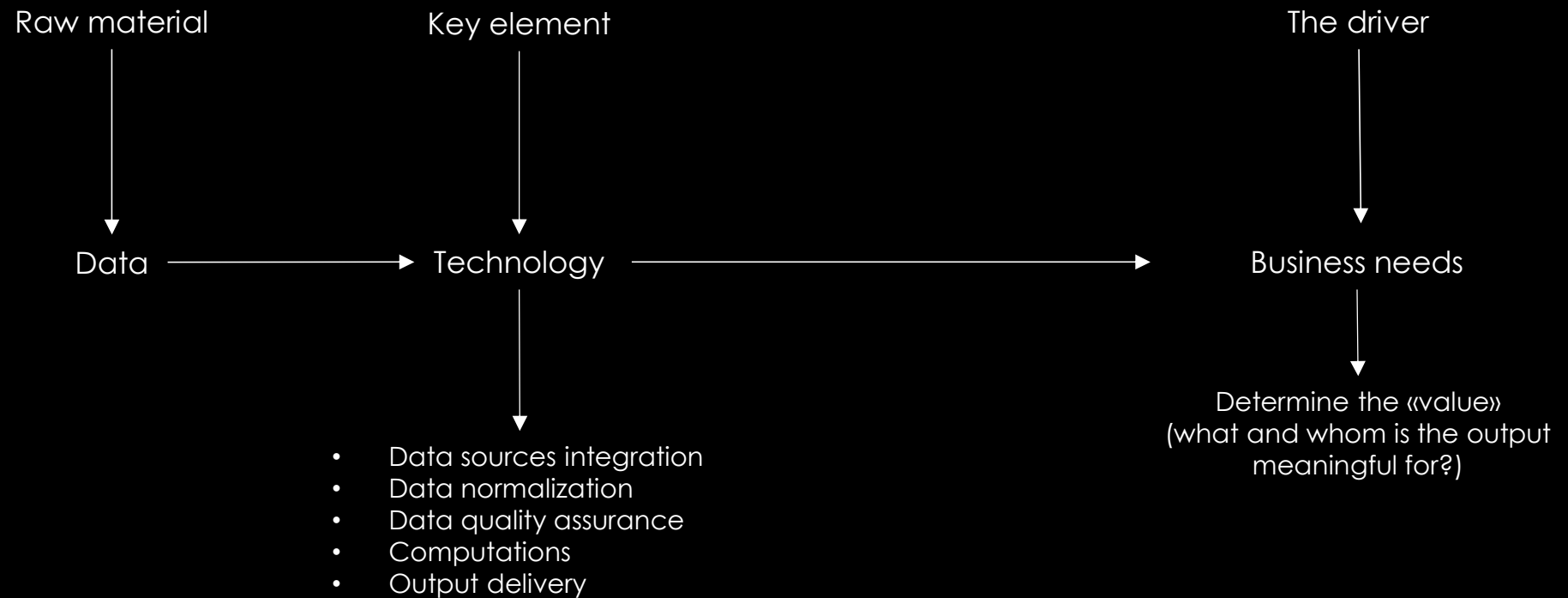
So the technological layer typical of data science, must «host» and implement the proper statistical methodologies otherwise, even if «convergent», it produces invalid knowledge.

The technological and the methodological layer must be tuned together.

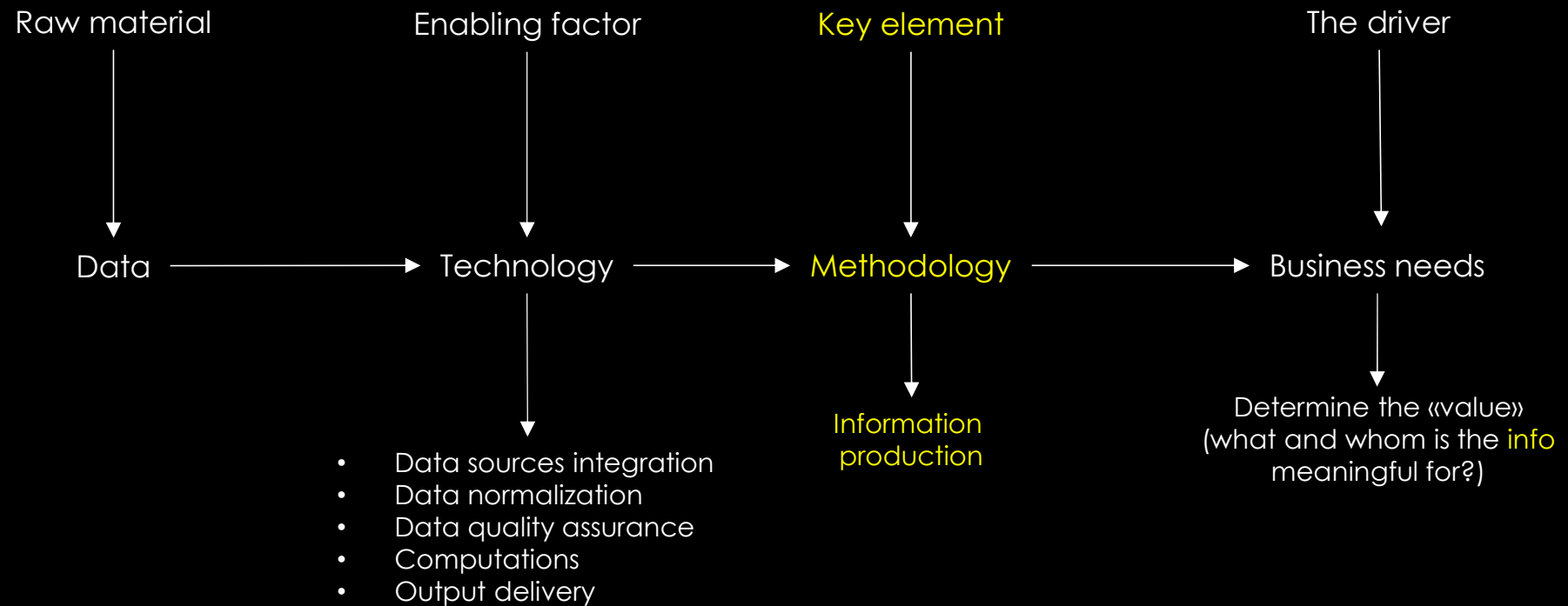
Unfortunately, this is seldom the case, as we see later.

The structure of data science processes

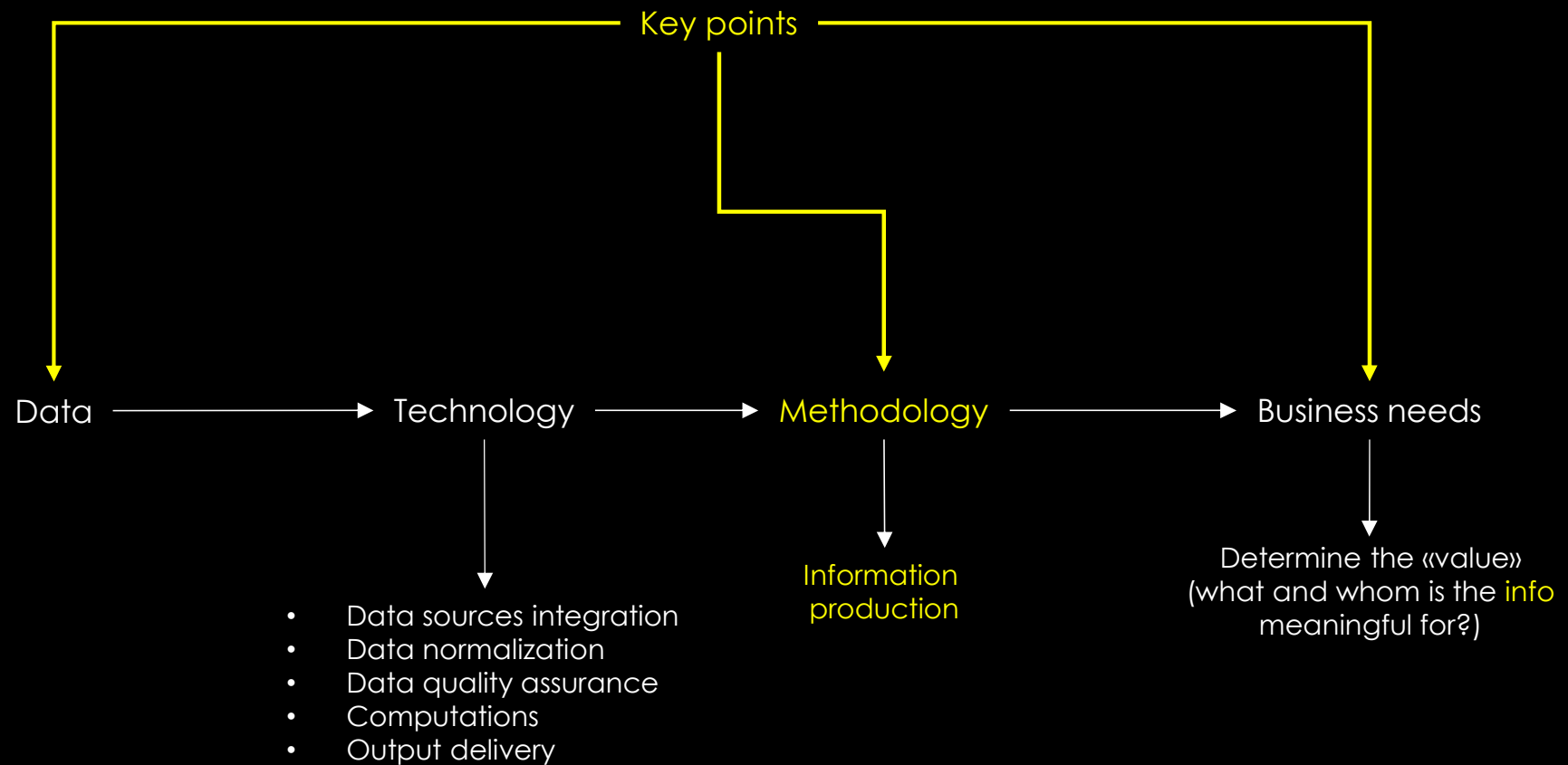
Value chain «old style»



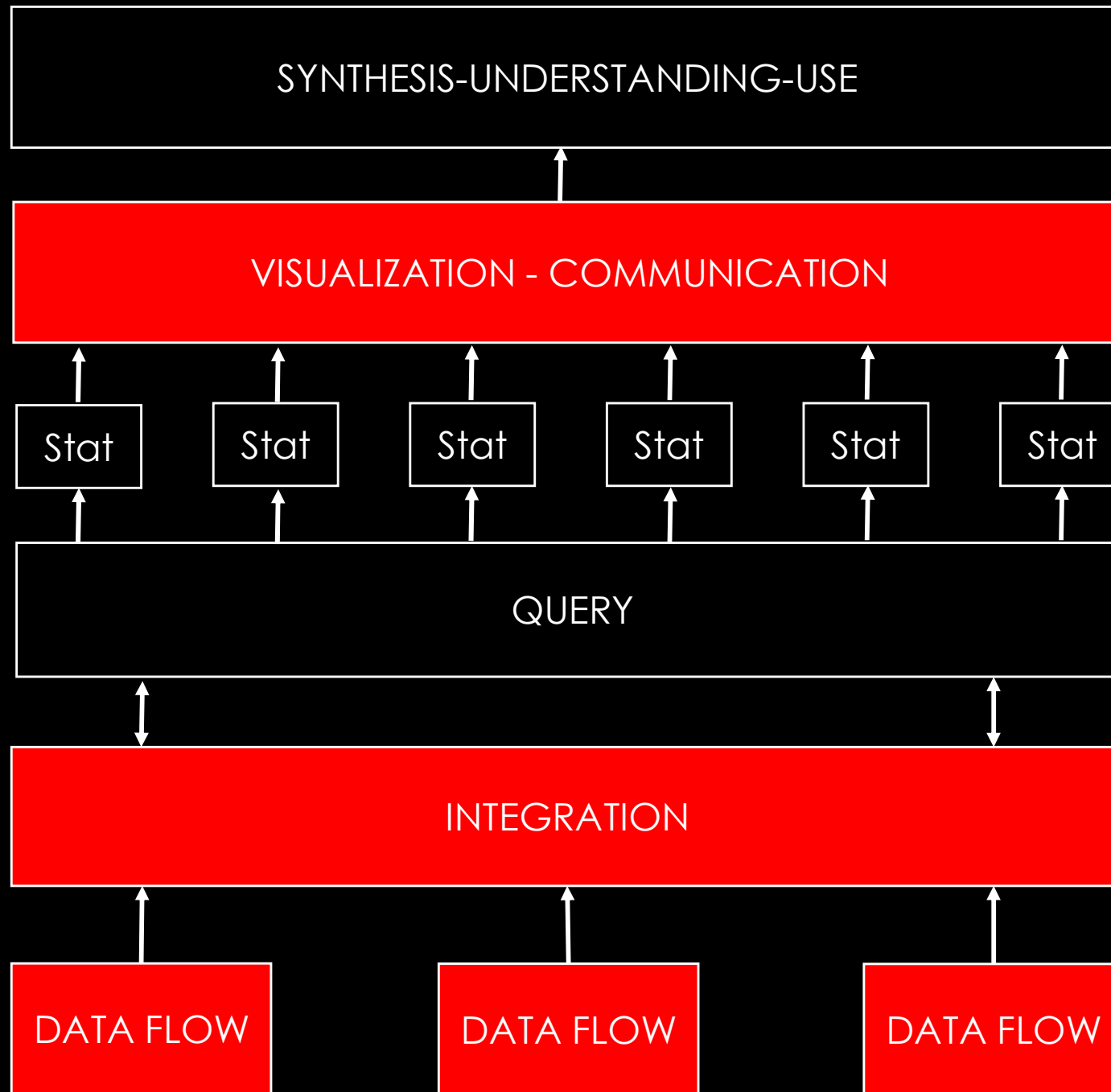
Value chain «new style»



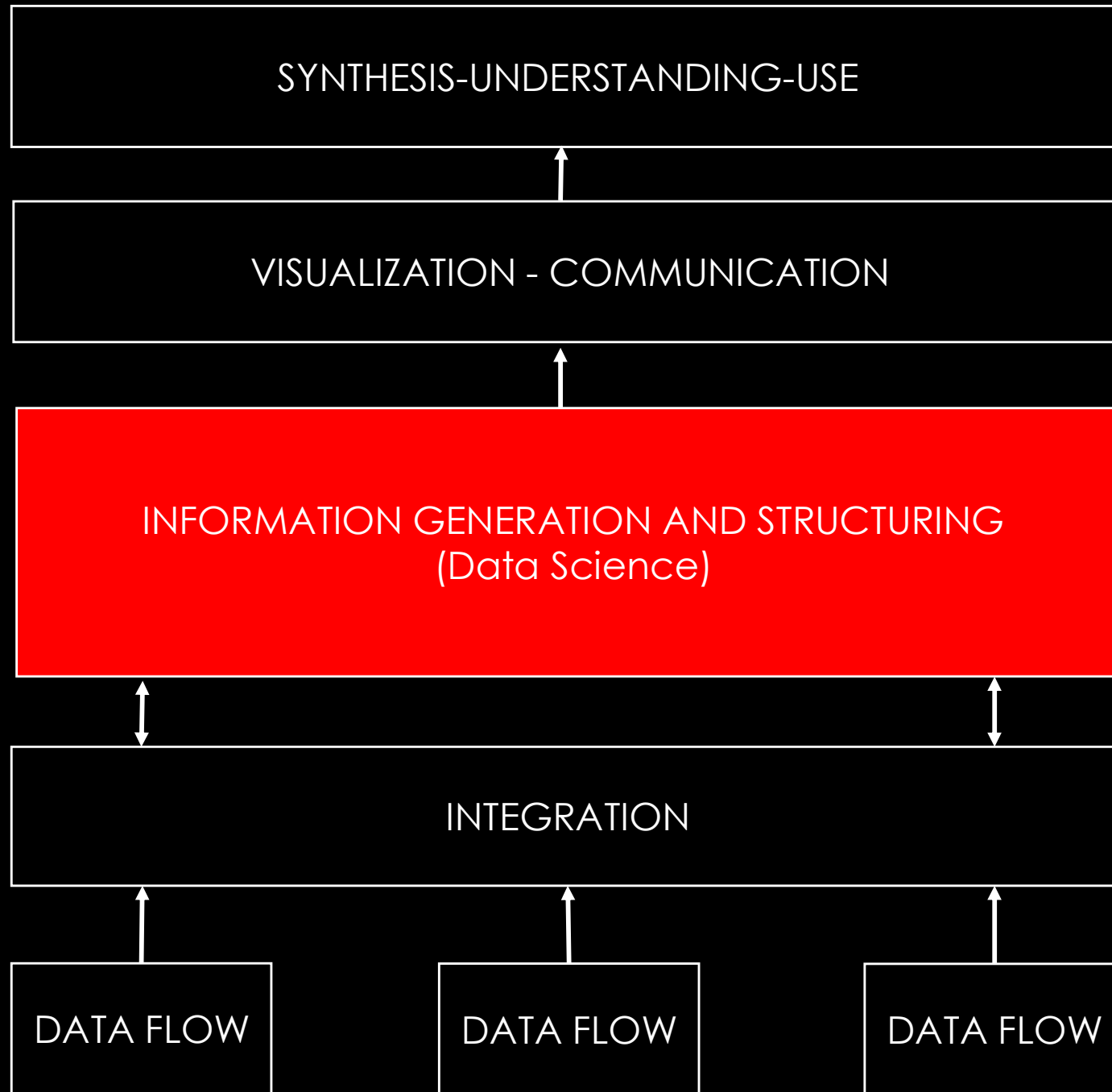
Value chain «new style»



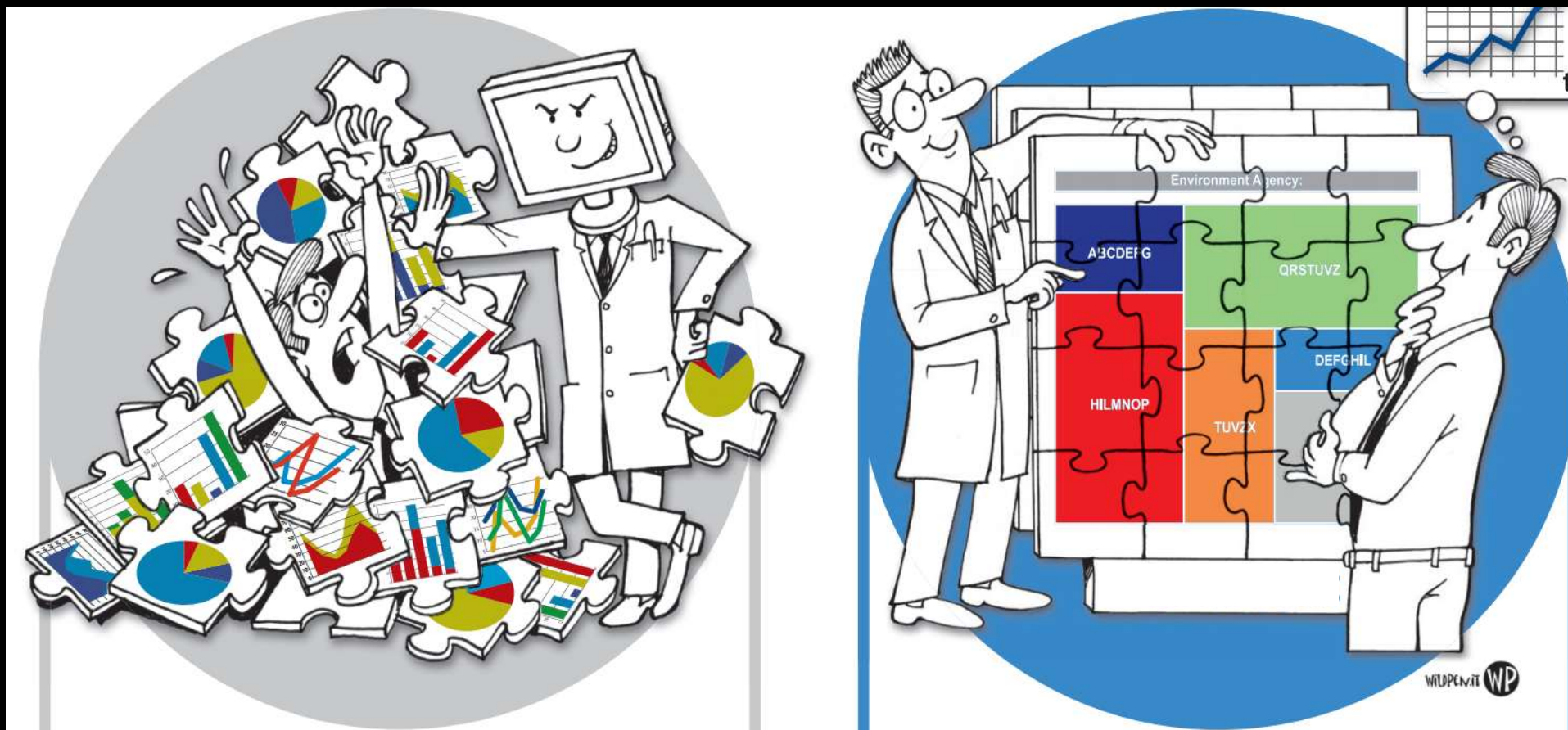
The classical model: informative process



The new model: informational process



In a picture



Example: the power of methodology

Market: advertising

Issue: customer targeting

Previous approach: an ad hoc and somehow arbitrary cluster analysis, identifying 10000 potential customers, on whom to concentrate the sales force.

New approach: customers have been partially ordered based on a set of «quality» indicators and the experience of managers. A ranking has been produced out of the partial order, leading to identifying 2000 true potential customers. Applying this approach to customers from the previous year, it has been checked that the 2000 identified customers actually produced 90% of the revenues.

Consequences: cost reduction and increased efficiency.

The structure of data science problems

Types of data science problems: conceptual map

QUESTION

Vague

Precise

Operational

Strategic

DATA

Heterogeneous

Homogeneous

Small

Big

COMPUTATIONS

Fast

Slow

Small

Massive

OUTPUT

To be interpreted

Immediate use

Simple

Complex

QUESTION

Vague		
Precise		
	Operational	Strategic

The diagram is a 2x2 matrix. The vertical axis is labeled 'Vague' at the top and 'Precise' at the bottom. The horizontal axis is labeled 'Operational' on the left and 'Strategic' on the right. The top-left cell (Vague, Operational) is empty. The top-right cell (Vague, Strategic) is empty. The bottom-left cell (Precise, Operational) is highlighted in red. The bottom-right cell (Precise, Strategic) is empty.

	Small	Big
Heterogeneous		
Homogeneous		

	Small	Massive
Fast		
Slow		

	Simple	Complex
To be interpreted		
Immediate use		

OUTPUT

Customer profiling (web data)

QUESTION

Vague		
Precise		
	Operational	Strategic

DATA

Heterogeneous		
Homogeneous		
	Small	Big

COMPUTATIONS

Fast		
Slow		
	Small	Massive

OUTPUT

To be interpreted		
Immediate use		
	Simple	Complex

Fraud detection

QUESTION

Vague		
Precise		
	Operational	Strategic

DATA

Heterogeneous		
Homogeneous		
	Small	Big

COMPUTATIONS

Fast		
Slow		
	Small	Massive

OUTPUT

To be interpreted		
Immediate use		
	Simple	Complex

Predictive maintenance

QUESTION

Vague		
Precise		
	Operational	Strategic

DATA

Heterogeneous		
Homogeneous		
	Small	Big

COMPUTATIONS

Fast		
Slow		
	Small	Massive

OUTPUT

To be interpreted		
Immediate use		
	Simple	Complex

The structure of data science projects

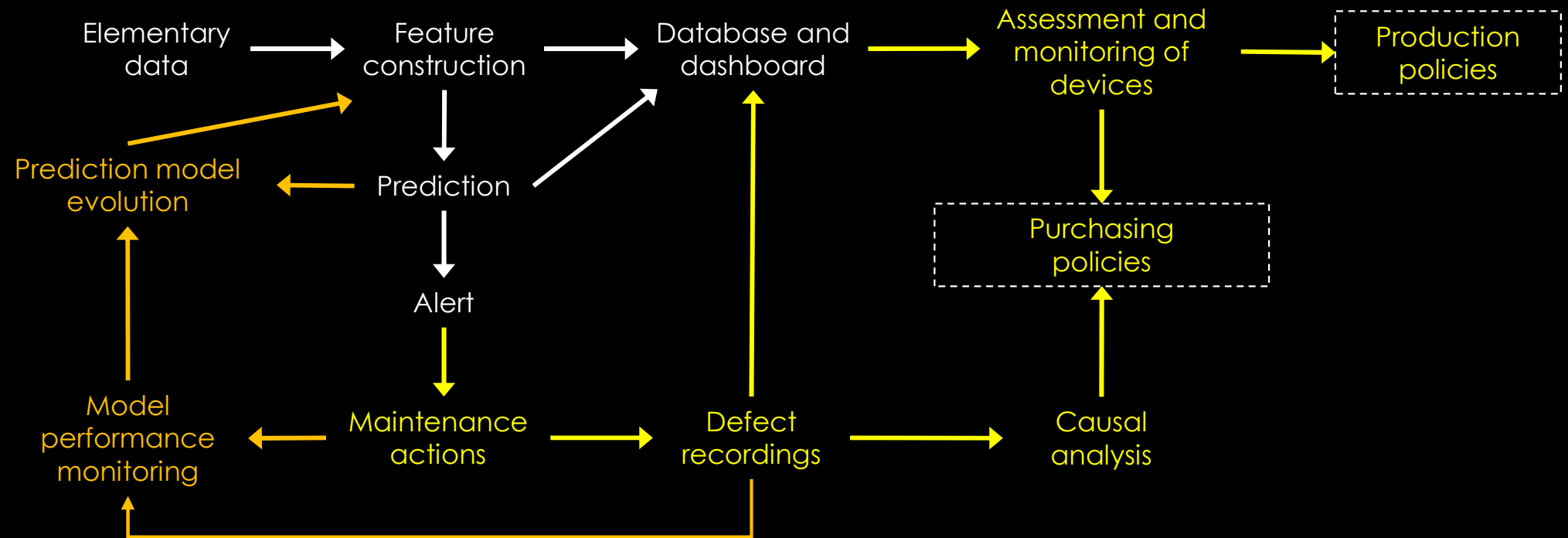
Informational engineering

As you design productive processes in an engineered way, so you must do for informational processes.

The processes interconnecting and transforming bits into information, and generating **informational layers**, requires the engineered integration of statistical methodologies and of these with technology.

So we need an **informational engineering**, to be put on top of software and technological engineering.

As any process, it must be monitored, assessed, validated, made evolve... so any of its component must be designed in order to facilitate these tasks.



Data science projects: structure

Phase 1 – Study and assessment



Phase 2a – Development and prototipization



Phase 2b - Implementation



Phase 1 – Study and assessment



Which malfunctionings do you want to identify?

Which maintenance actions do you want to anticipate?

What does it mean to «read the market»?

What does it mean to «sell more»?

Which is a «good behavior» by a user?

What does it mean to «optimize the budget»?

Who will use the service?

Which infos are relevant to the users?

How users access the infos?

How frequently must infos be updated?

How are data generated?

Which variables are produced and how?

Data quality? (syntactical)

Data quality? (semantic)?

Do data comprise the relevant elements to answer the question/business need?

Phase 2a – Development and prototyping



Which output must be delivered?

Which features must be recorded?

Device must be ordered in terms of malfunctioning probability? or severity?

Environment setting and NDA

Definition of the statistical criteria for data sampling and acquisition

Data sampling and storing.

Identification of possible models and algorithms

Identification of quality criteria for the output information

Identification of possible performance constraints (e.g. computation time, memory space, storing costs...)

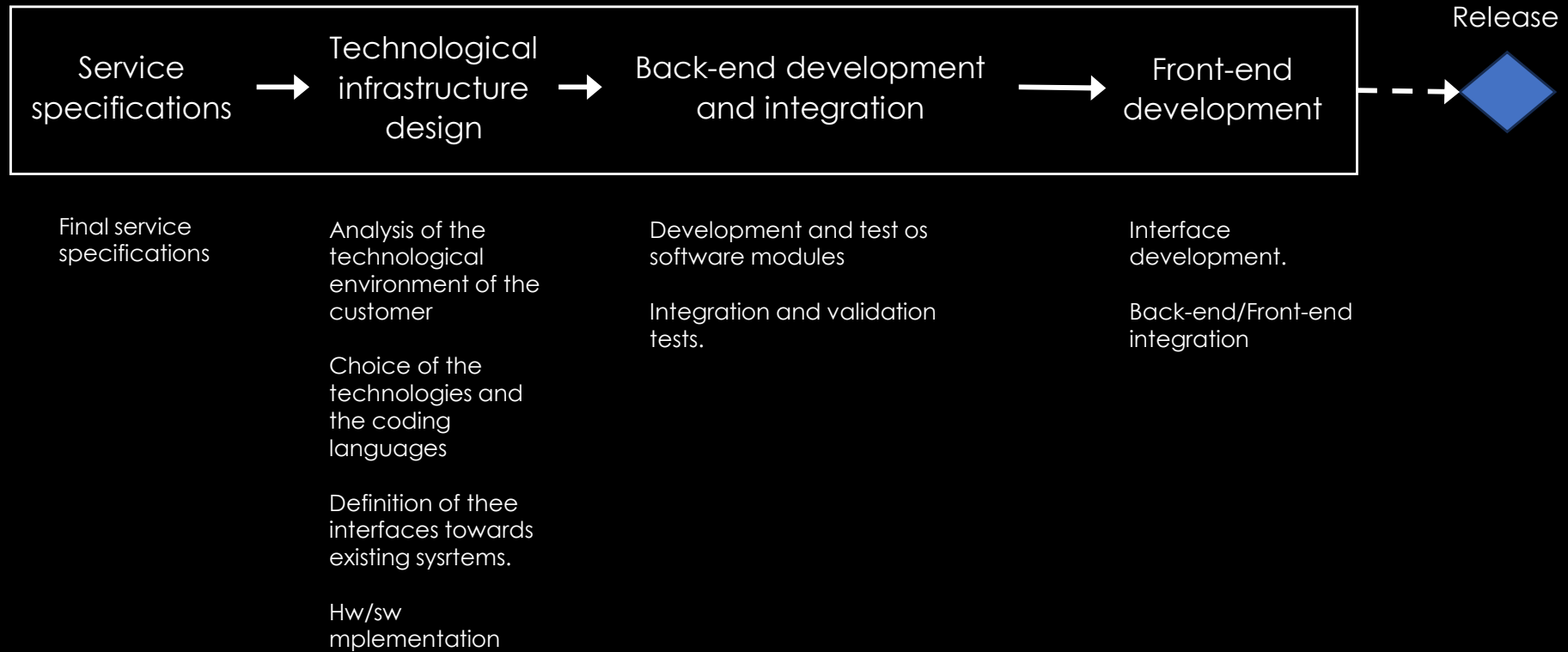
Development, test and model comparison

Model selection

Prototyping

Interface sketch

Phase 2b - Implementation



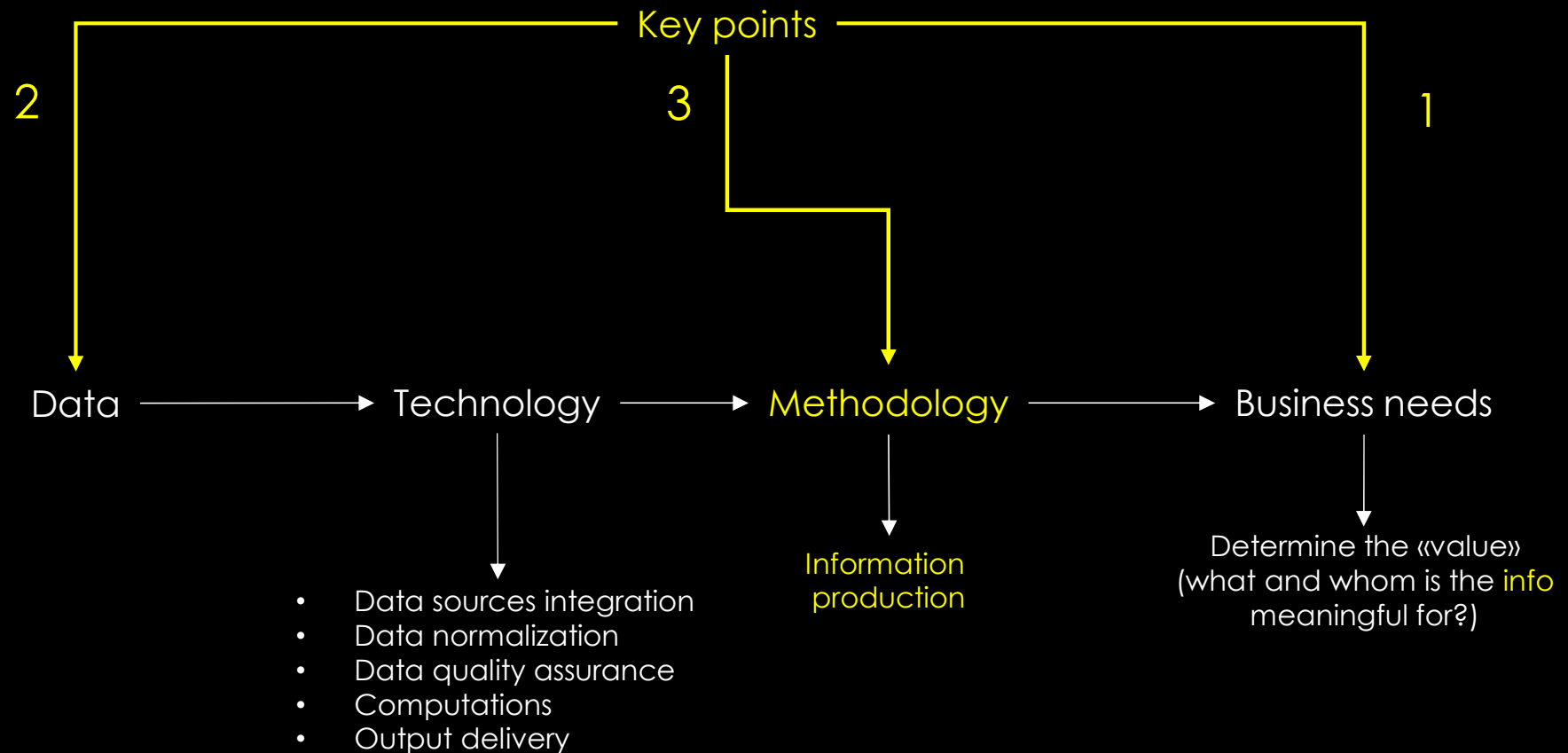
The key ingredients

Typical students' approach (see dslab projects)

The prediction performances of our model are not good enough, so we need more effective algorithms

Are algorithms the true problem?

Value chain «new style»



1 – Conceptualizing the question

Strange as it may seem, the first and most problematic issues in data science projects is the vagueness of the questions.

Examples:

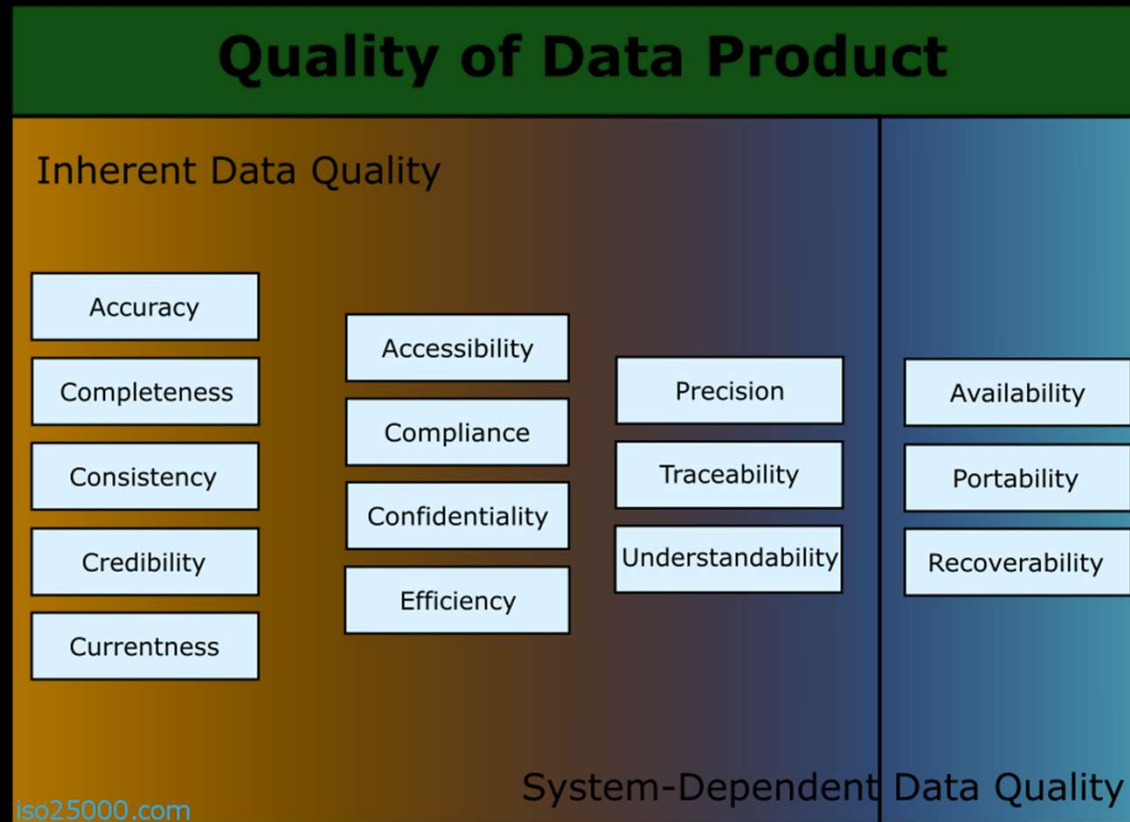
1. Cinema tickets
2. What is the value of a product? and of a customer?
3. What is well-being?
4. What is the quality of a service?

We will deepen this topic later, when talking about measurement.

2 – The power (or non-power) of data

No algorithm can extract information, unless this is already comprised in the input data.

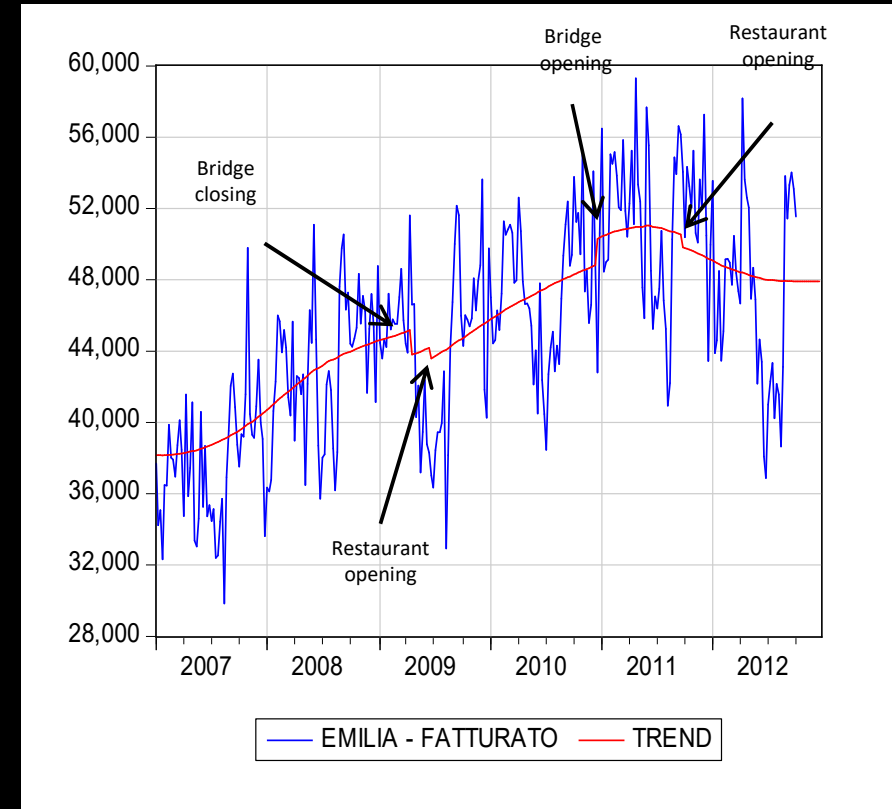
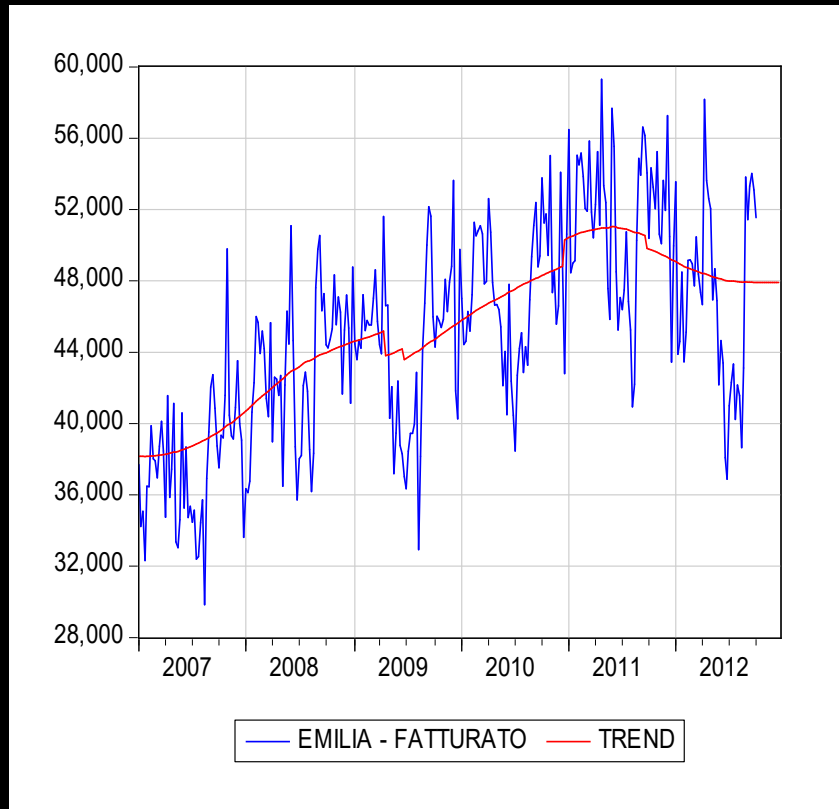
1. Cinema tickets
2. Budget optimization: tv
3. E-commerce - customer preferences
4. Agents effectiveness
5. Power absorption sampling
6. Customer satisfaction survey



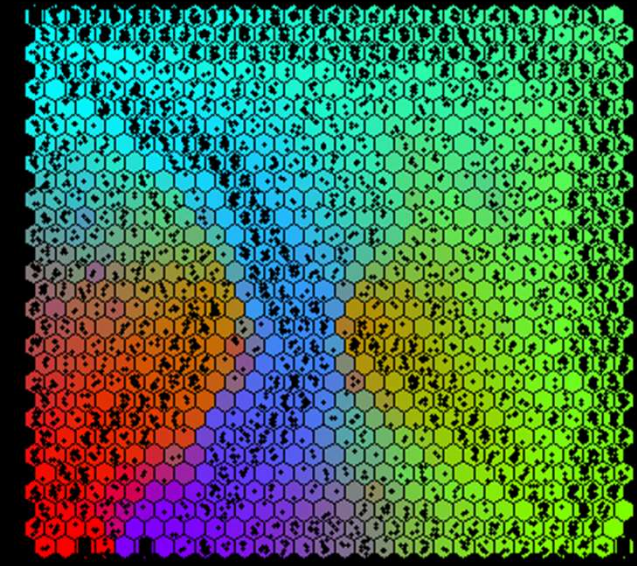
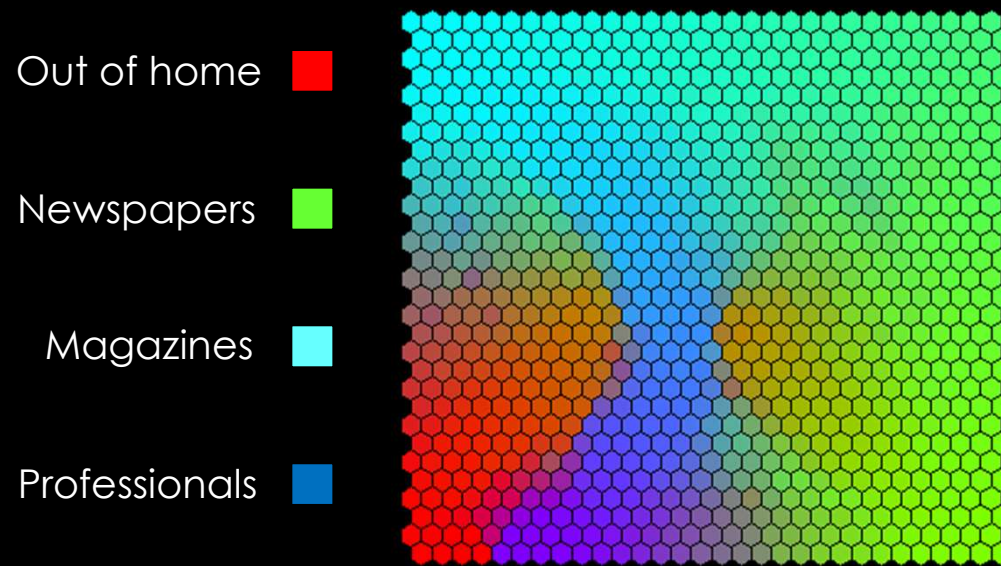
Is this enough?

Data quality: the capability of data to «support» the knowledge generation process.

3 - Methodology



3 - Methodology



3 - Methodology

